L2

Linux

Scenario-Based interview Ques. & short Ans.

By Akash Nimje

https://www.linkedin.com/in/akash-nimje-4065b0231/

L2- Linux Scenario-Based interview question & short Ans strong, practical answers.

L2 Scenario-Based Linux Interview Questions with Answers

1) Load average high, CPU normal

- Check I/O wait, blocked processes, disk latency
- Tools: `top`, `iostat`, `vmstat`, `ps -eo stat`

2) SSH access lost

- Use cloud/VM console or KVM
- Check disk, fs, SSH config, firewall
- Avoid reboot unless required

3) Emergency mode after power failure

- Check `/etc/fstab`, fs corruption
- Run `fsck`, fix UUIDs
- Remount root RW and reboot

4) App timeout but ping works

- Check ports, firewall, app threads
- Tools: `ss`, `netstat`, `telnet`, logs

5) Memory usage keeps growing

- Monitor per-process memory over time
- Check cache vs RSS
- Use `smem`, `pmap`, app logs

6) Intermittent packet loss

- Check NIC errors, drops, MTU
- Tools: `ethtool`, `sar -n`, `tcpdump`

7) High disk I/O latency

- Identify disk vs process issue
- Tools: `iostat -x`, `iotop`, `df`, `dmesg`

8) Works in staging, fails in prod

- Compare configs, env vars, SELinux
- Check permissions and dependencies

9) Frequent OOM killer

- Identify memory-hungry processes
- Check swap, limits, leaks
- Tune or scale memory

10) High CPU, no single process

- Check soft IRQs, kernel threads
- Look for context switching
- `vmstat`, `mpstat`

11) Random reboots

- Check hardware logs, kernel panic
- Review `journalctl`, IPMI, dmesg

12) Kernel panic RCA

- Capture crash logs, version, workload
- Check recent changes
- Analyze core dump if available

13) App fails DB connection intermittently

- Check network, DNS, connection pool
- Validate DB max connections

14) NFS data inconsistency

- Check mount options, sync/async
- Verify server and client logs

15) Time drift issues

- Verify NTP/chrony
- Check firewall and RTC
- Force sync and persist config

16) Disk slow after filesystem extend

- Fragmentation or snapshot overhead
- Recheck alignment and I/O scheduler

17) Slow login during peak hours

- Check LDAP, PAM, DNS latency
- Review auth logs and CPU spikes

18) Disk fills nightly

- Find growing files
- `du -xh`, logs, backups, cron jobs

19) Low network throughput

- Check MTU mismatch, NIC speed
- Test with `iperf`, `ethtool`

20) Service crashes after running

- Memory leak or resource exhaustion
- Review logs, limits, core dumps

21) System hangs but ping works

- Kernel alive, userspace blocked
- Check load, I/O wait, fs locks

22) High context switching

- Too many threads or I/O wait
- Indicates scheduler pressure

23) Random permission errors

- SELinux contexts
- NFS root squash
- ACL issues

24) Performance drop after LVM snapshot

- Snapshot COW overhead
- Remove or resize snapshot

25) Intermittent auth failures

- PAM, LDAP, DNS, time sync
- Correlate logs across layers

26) Kernel upgrade boot failure

- Boot older kernel via GRUB
- Rollback and rebuild initramfs

27) Inode exhaustion

- `df -i`
- Clean small files, logs, caches

28) Mounted FS but data missing

- Mounted wrong device
- Overlay or bind mount issue

29) Logs missing after reboot

- Volatile journald storage
- Check `/var/log`, disk space

30) App fails under heavy load

- Race conditions, limits, scaling
- Load testing + tracing

31) NFS mount hangs

- Network latency, server load
- Check `soft/hard` mount option

32) Overlapping cron jobs

- Add locking (`flock`)
- Redesign schedule

33) Internal network OK, external down

- Routing, NAT, firewall
- Check default gateway

34) High swap despite free RAM

- Kernel prefers idle pages
- Check swappiness

35) Zombie processes

- Parent not reaping children
- Fix application logic

36) Performance drop after patching

- Compare before/after metrics
- Rollback or tune

37) Disk I/O errors, SMART OK

- Check cables, controller, logs
- `dmesg`, RAID status

38) Cannot bind to port

- Port already in use
- TIME_WAIT sockets
- `ss -lntp`

39) Emergency mode recovery

- Same approach: fs, fstab, UUID
- Manual fix, reboot

40) High file descriptor usage

- Leaking connections
- Increase limits and fix app

41) One app slow disk writes

- Sync writes, fs options
- App-level flush behavior

42) One-direction latency

- Duplex mismatch
- Routing or firewall asymmetry

43) Permission denied but perms OK

- SELinux
- Mount options (noexec, nosuid)

44) Dependency failure in services

- Fix `After`, `Requires`
- Validate startup order

45) Kernel upgrade breaks network

- Boot old kernel
- Rebuild drivers/modules

46) System hangs on shutdown

- Stuck services or NFS
- Check shutdown logs

47) df shows full after cleanup

- Deleted open files
- Restart process holding FD

48) App behaves differently after reboot

- Env vars, cache, tmp files
- Compare runtime state

49) Config drift outage

- Use config management
- Version control and audits

50) Multiple services fail after change

- Timeline analysis
- Identify common dependency
- Rollback and RCA

@Akash Nimje
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